

🔥 XAI: Probing Further into 'Interpretability': Caveats & Challenges

FAMAF - UNC

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eXplainable
Artificial Intelligence

Agenda ---

- Two Papers:
 - ▶ Transparency: Motivations and Challenges
 - ▶ The Mythos of Model Interpretability
- Break out groups
- Discussion

🍲 The Mythos of Model Interpretability ---

Zachary Lipton; 2017

Research Focus

Contributions ---

- **Goal:** Refine the discourse on interpretability
- Outline desiderata of interpretability research
- Motivations for interpretability are often diverse and discordant
- Identifying model properties and techniques thought to confer interpretability

Motivation ---

- We want models to be not only good w.r.t. predictive capabilities, but also **interpretable**
- Interpretation is underspecified
 - ▶ Lack of a formal technical meaning
 - ▶ Papers provide diverse and non-overlapping motivations for interpretability

🍷 Prior Work: Motivations for Interpretability ---

Interpretability promotes trust

- But what is trust?
- Is it faith in model performance?
- If so, why are accuracy and other standard performance evaluation techniques inadequate?

🍷 When is interpretability needed? ---

- Simplified optimization objectives fail to capture complex real life goals
 - ▶ Algorithm for hiring decisions – productivity **and** ethics
 - ▶ Ethics is hard to formulate
- Training data is not representative of deployment environment
Interpretability serves those objectives that we deem important but struggle to model formally!

Desiderata ---

Understanding motivations for interpretability through the lens of prior literature:

- **Trust**
- **Causality**
- **Informativeness**
- **Fair and Ethical Decision Making**

Desiderata: Trust ---

- Is trust simply confidence that the model will perform well?
- A person might feel at ease with a well understood model, even if this understanding has no purpose
- Training and deployment objectives diverge
 - ▶ E.g., model makes accurate predictions but not validated for racial biases
- Trust → relinquish control
- For which examples is the model right?

Desiderata: Causality ---

- Researchers hope to infer properties (beyond correlational associations) from interpretations/explanations
- Regression reveals strong association between smoking and lung cancer
- However, task of inferring causal relationships from observational data is a field in itself
 - ▶ Don Rubin
 - ▶ Judea Pearl

🍷 Desiderata: Informativeness ---

- Predictions $\rightarrow$ Decisions
- Convey additional information to human decision makers
- **Example:** Which conference should I target?
 - ▶ A one word answer is not very meaningful
- Interpretation might be meaningful even if it does not shed light on model's inner workings
- Similar cases for a doctor in support of a diagnosis

🍲 Desiderata: Fair & Ethical Decision Making ---

- ML is being deployed in critical settings
 - ▶ E.g., healthcare
- How can we be sure algorithms do not discriminate on the basis of race?
- AUC is not good enough
- **Side note:** European Union – Right to explanation

Properties of Interpretable Models ---

Transparency

- How exactly does the model work?
- Details about its inner workings, parameters etc.

Post-hoc explanations

- What else can the model tell me?
- E.g., visualizations of learned model, explaining by example

🍷 Transparency: Simulatability ---

- Can a person contemplate the entire model at once?
- Need a very simple model
- A human should be able to take input data and model parameters and calculate prediction

🍷 Transparency: Decomposability ---

- Understanding each input, parameter, calculation
- E.g., decision trees, linear regression
- Inputs must be interpretable
- Models with highly engineered or anonymous features are not decomposable

🍷 Algorithmic Transparency ---

- Learning algorithm itself is transparent
- E.g., linear models (error surface, unique solution)
- Modern deep learning methods lack this kind of transparency
 - ▶ We don't understand how the optimization methods work
 - ▶ No guarantees of working on new problems
- **Note:** Humans do not exhibit any of these forms of transparency

🍷 Post-hoc: Text Explanations ---

- Humans often justify decisions verbally (post-hoc)
- **Krening et. al.:**
 - ▶ One model is a reinforcement learner
 - ▶ Another model maps models states onto verbal explanations
 - ▶ Explanations are trained to maximize likelihood of ground truth explanations from human players
- **So:** explanations do not faithfully describe agent decisions, but rather human intuition

Post-hoc: Visualization ---

- Visualize high-dimensional data with t-SNE
 - ▶ 2D visualizations in which nearby data points appear close
- Perturb input data to enhance activations of certain nodes in neural nets (image classification)
 - ▶ Helps understand which nodes corresponds to what aspects of the image
 - ▶ E.g., certain nodes might correspond to dog faces

🍷 Post-hoc: Example Explanations ---

- Reasoning with examples
- E.g., Patient A has a tumor because he is similar to these k other data points with tumors
- k neighbors can be computed by using some distance metric on learned representations
- E.g., word2vec

🍲 Post-hoc: Local Explanations ---

- Hard to explain a complex model in its entirety
- How about explaining smaller regions?
- Explains decisions of any model in a local region around a particular point
- Learns sparse linear model

LIME (Ribeiro et. al.)

LIME Visualization

🍷 Key Insights ---

- Claims about interpretability must be qualified
 - ▶ If a model satisfies a form of transparency, highlight that clearly
 - ▶ For post-hoc interpretability, fix a clear objective and demonstrate evidence

Potential Issues ---

Transparency may be at odds with broader objectives of AI

- Choosing interpretable models over accurate ones to convince decision makers
- Short term goal of building trust with doctors might clash with long term goal of improving health care



Post-hoc interpretations can mislead ---

- Do not blindly embrace post-hoc explanations!
- Post-hoc explanations can seem plausible but be misleading
- They do not claim to open up the black-box
- They only provide plausible explanations for its behavior
- E.g., text explanations

🍷 Summary: The Mythos of Model Interpretability ---

- **Goal:** Refine the discourse on interpretability
- Outline desiderata of interpretability research
- Motivations for interpretability are often diverse and discordant
- Identifying model properties and techniques thought to confer interpretability

🍷 Transparency: Challenges and Motivation ---

Adrian Weller; 2019

Research Focus

Contributions ---

- Characterizing different kinds of transparency, and underlying motivations
- Shedding light on the downsides of having transparency

🍲 Types and Goals of Transparency ---

Type 1: For a developer, to understand how their system is working, aiming to debug or improve it: to see what is working well or badly, and get a sense for why.

Type 2: For a user, to provide a sense for what the system is doing and why, to enable prediction of what it might do in unforeseen circumstances and build a sense of trust in the technology.

Type 3: For society broadly to understand and become comfortable with the strengths and limitations of the system, overcoming a reasonable fear of the unknown.

🍲 Types and Goals of Transparency (cont.) ---

Type 4: For a user to understand why one particular prediction or decision was reached, to allow a check that the system worked appropriately and to enable meaningful challenge (e.g. credit approval or criminal sentencing).

Type 5: To provide an expert (perhaps a regulator) the ability to audit a prediction or decision trail in detail, particularly if something goes wrong (e.g. a crash by an autonomous car). This may require storing key data streams and tracing through each logical step, and will facilitate assignment of accountability and legal liability.

🍲 Types and Goals of Transparency (cont.) ---

Type 6: To facilitate monitoring and testing for safety standards.

Type 7: To make a user (the audience) feel comfortable with a prediction or decision so that they keep using the system. Beneficiary: deployer.

Type 8: To lead a user (the audience) into some action or behavior – e.g. Amazon might recommend a product, providing an explanation in order that you will then click through to make a purchase. Beneficiary: deployer.

Global vs. Local ---

Global

- Understanding whole system
- Types 2 – 3

Local

- Explanation for a particular prediction
- Types 4, 5, 7, 8

🍷 Key Challenges ---

- Explanations are beneficial to the society only if they are **faithful**
- Notion of faithfulness is hard to characterize precisely!
- Defining criteria and tests for practical faithfulness are important open problems
- **Context is important!**

🍷 Understanding and Communication ---

- **Challenge:** Is an explanation good at conveying faithful information in understandable form, and if a human has actually understood it well?
- Context dependent
- Need for deeper probing!

🍷 Comparing Explanations ---

If we have two different saliency maps, how to know which one is better

Saliency Maps Comparison (*Two images showing different saliency maps for the same cat and dog photo*)

🍷 Possible Dangers: Audience vs. Beneficiary ---

- **Recommender systems**

- ▶ Amazon (Beneficiary) and its Users (Audience)

- **Healthcare**

- ▶ Google Verily (Beneficiary) and Hospitals/Doctors (Audience)

- **Criminal Justice**

- ▶ COMPAS (Beneficiary) and Courts/Judges (Audience)

🍷 Government Use of Algorithms ---

- COMPAS system predicts risk of recidivism
- A prisoner should have some transparency into the decision made by COMPAS
 - ▶ To ensure proper process has been followed
 - ▶ Enable potential challenge
- **But:** can there be too much transparency?
- Also, recent push for making all code/data for such models public. Good idea?

🍷 Gaming, IP Incentives, and Privacy ---

- If all details available, the process can be **gamed**
- Less incentive for private IP and slow progress
- Privacy and transparency are often in **conflict**
- How much transparency is too much in a setting?

🍷 Means and Ends ---

- Transparency → reliability, fairness
- If we are able to develop a good set of “safety checks”, then may be it is ok to not have full transparency
- E.g., autonomous vehicles

Societal Considerations ---

Does giving more information to each individual help society?

- **Braess' paradox:** more information empowers the agents to optimize their own agendas more efficiently, and thus may lead to a worse global outcome

🍷 Selective Transparency & Discrimination ---

- Selective Transparency may hurt people disproportionately
- Furthermore, transparency about certain attributes (e.g., gender) in certain settings is known to cause discriminatory behavior as well

Break Out Groups ---

Based on the different papers we have covered, can you summarize your stance on:

- When/Why is interpretability needed?
- When is interpretability a bad idea?
- Are there any privacy concerns around making models interpretable? What about fairness concerns?
- In real world settings, there have been cases where simpler models were chosen over accurate ones to secure “trust” of decision makers. What do you think about this?